

# SPHERE Thermal Insulation Kit

## Team Engineering Report

Piyath Vithanage, Elio Krollpfeiffer, Dilan Fernando, and Nuhansee Migelhewage

TUM School of Engineering and Design  
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**Abstract**—We designed a classroom experiment that compares four insulation arrangements around a 100 mL warm-water core. Students record 16 temperatures, generate a Newtonian cooling reference curve, and evaluate the difference between measurement and prediction. The worked test ranked the full cotton–bubble-wrap–Mylar package first for heat retention. The report also explains why the water-bath test is not a vacuum or spacesuit qualification test.

**Keywords**—heat transfer, Newtonian cooling, insulation, classroom experiment, data logging, model validation

This report presents our SPHERE thermal-insulation classroom kit. We built the project around one engineering question: how much does a chosen layer arrangement reduce the cooling rate of the same 100 mL water-filled core under the same cold-water boundary? The kit combines four physical test articles, a browser-based Mission Control page, manual temperature logging, and a Newtonian cooling model. Our aim was to make heat transfer measurable in one school lesson without requiring specialist data-acquisition hardware.

The four conditions are a bare control, two layers of bubble wrap, a Mylar-only wrap, and a full multilayer package made from cotton, bubble wrap, and Mylar. Students record the core temperature once per minute from 0 to 15 minutes. The website plots the recorded values against a reference curve and calculates residuals and mean absolute error. In the worked data, the full multilayer package retained the most heat. Mylar alone performed poorly in the water-bath setup, which showed why a reflective material cannot be judged without considering contact, convection, wetting, gaps, and closure quality.

### Main outcome

We produced a reusable experiment that links construction, measurement, modelling, and evaluation. The activity does not reproduce vacuum and does not test materials for real spacesuit use. It is a controlled teaching model for comparing cooling behaviour. The final design uses identical glass jars, a target start temperature near 80 °C, a measured ice-water bath, a centred probe, and a 16-point temperature record.

| Item          | Final project setting                       |
|---------------|---------------------------------------------|
| Thermal core  | 100 mL water in matched glass jars          |
| Boundary      | Measured ice-water bath                     |
| Sampling      | Every minute from 0 to 15 min               |
| Outputs       | Measured curve, model curve, residuals, MAE |
| Lesson length | About 55-70 min including setup and review  |

### Team scope

The documented task split assigns the website and graphing workflow to Piyath Vithanage. It assigns the thermal model, experiment, analysis, safety, documentation, and integration to Dilan Fernando. Elio Krollpfeiffer and Nuhansee Migelhewage are listed as team members, but the source materials do not identify their specific deliverables. Their individual statements therefore require their confirmation before submission.

## 1 Project Context and Requirements

### 1.1 Context

SPHERE uses human-spaceflight engineering as the setting for classroom activities. We chose thermal protection because students can see the construction, measure a temperature change, and compare a real curve with a mathematical prediction. The spacesuit story gives the task a clear purpose, but the measured system is a jar in a water bath. We kept that distinction visible throughout the manuals and website.

### 1.2 Engineering requirements

- Produce a visible difference between configurations within a 15-minute test.
- Use parts that a school can replace locally.
- Keep the procedure understandable without prior differential-equation study.
- Run on ordinary laptops and tablets without accounts, pairing, or a native app.
- Keep hot water, cold bath water, and electronic displays separated.
- Use the same variables and test settings in the website, manuals, worksheet, and report.

### 1.3 Learning requirements

Students should be able to distinguish conduction, convection, and radiation; explain the meaning of an effective cooling constant; collect an ordered time series; compare measurement with prediction; and discuss uncertainty. We also wanted them to see that an unexpected result is useful evidence. A disagreement should lead to checks of the setup and assumptions, not to editing the data until it matches the model.

| Requirement         | Design response          | Check                    |
|---------------------|--------------------------|--------------------------|
| Comparable cores    | Matched 100 mL jars      | Same geometry and fill   |
| Repeatable boundary | Shared ice-water bath    | Measure bath temperature |
| Safe handling       | Adult pours hot water    | Pre-run safety checklist |
| Accessible software | Static browser app       | No login or driver       |
| Traceable result    | 16 raw readings retained | Table and graph agree    |

### 1.4 Decisions made during development

We first considered automatic sensors and wireless transfer. We rejected that option because it would shift attention from the thermal problem to pairing, drivers, and battery faults. Manual entry makes the observation step explicit and is easier for a teacher to recover when something goes wrong. We also changed the design question from a general spacesuit demonstration to a controlled comparison of cooling rate. That change gave us measurable inputs, a fair-test procedure, and a result that students could defend.

## 2 Thermodynamic Model

### 2.1 Lumped cooling model

We model the water as one thermal mass with a uniform temperature. This is an approximation, but it gives a useful reference for a short classroom test. The stored sensible energy change is

$$Q = m c_p \Delta T.$$

For 100 mL of water, the mass is approximately 0.10 kg. With a water heat capacity of about  $4184 \text{ J kg}^{-1} \text{ K}^{-1}$ , a 10 K drop represents about 4.18 kJ leaving the water. The jar, probe, and insulation also store energy, so this value is an estimate rather than a complete energy balance.

For a body losing heat to an environment at approximately constant temperature, we use

$$m c_p dT/dt = -hA(T - T_{env}),$$

which gives the reference curve

$$T(t) = T_{env} + (T_o - T_{env})e^{-kt}.$$

The website uses minutes, so the effective cooling constant  $k$  is in  $\text{min}^{-1}$ . It combines geometry, boundary conditions, thermal mass, and all heat-loss paths. It is not the material conductivity of cotton, bubble wrap, or Mylar.

### 2.2 Model coefficients

| Condition    | Reference $k$ ( $\text{min}^{-1}$ ) | Model T15 for 80/0 °C |
|--------------|-------------------------------------|-----------------------|
| Bare control | 0.150                               | 8.4 °C                |
| Bubble wrap  | 0.040                               | 43.9 °C               |
| Mylar only   | 0.121                               | 13.0 °C               |
| Full MLI     | 0.015                               | 63.9 °C               |

### 2.3 Interpretation

Cotton and bubble wrap reduce direct solid contact and trap low-density regions. Mylar mainly changes radiative exchange when a suitable air gap exists. In a water bath, conduction and convection can dominate, especially if water touches the wrap or enters through the neck. This is why the Mylar-only condition should not be treated as a direct test of spacecraft multilayer insulation.

For runs with different starting temperatures, we compare the dimensionless temperature above the boundary:

$$\theta(t) = [T(t) - T_{env}] / [T_o - T_{env}].$$

### 3 Hardware and Test Configurations

#### 3.1 Apparatus

The physical system uses four matched 100 mL glass jars, four digital probe thermometers, a separate bath thermometer, one stable bucket or deep container, insulation materials, a timer, towels, and heat-resistant handling tools. Each probe is suspended near the centre of the water volume. The display and cable connector stay dry and outside the bath.

| Condition    | Construction                         | Purpose                    |
|--------------|--------------------------------------|----------------------------|
| Bare control | Jar with no added insulation         | Reference heat-loss path   |
| Bubble wrap  | Two even layers with cells uncrushed | Trapped-air layer          |
| Mylar only   | One reflective layer, neck closed    | Reflective layer by itself |
| Full MLI     | Cotton, bubble wrap, outer Mylar     | Combined mechanisms        |

#### 3.2 Construction controls

Small assembly differences changed the response during development. A probe touching the glass did not represent the bulk water. Tight tape flattened bubble cells. A loose opening around the probe created a direct heat-loss path and could let bath water enter. We therefore treated probe position, layer compression, closure, and immersion depth as controlled variables.

- Use jars with the same diameter, wall thickness, and fill height.
- Cut insulation pieces to the same nominal size for repeated trials.
- Secure each layer only tightly enough to stop movement.
- Keep the jar neck and probe opening as consistent as possible.
- Record any wetting, gap, leak, or probe movement with the run data.

#### 3.3 Full multilayer package

The full package is built from the core outward: cotton next to the jar, two layers of bubble wrap around the cotton, and Mylar on the outside. The order was chosen so the soft inner layer follows the jar, the bubble layer provides trapped air, and the reflective sheet forms the outer surface. This arrangement is a classroom analogue only. Real spacecraft multilayer insulation uses carefully controlled low-emittance films, spacers, vacuum conditions, and qualification tests.

#### 3.4 Integration with the learning kit

The hardware labels match the configuration names in Mission Control. This prevents a group from choosing one model coefficient while testing another construction. The same four names appear in the manuals, worksheet, result table, and assessment questions.

## 4 Experimental Method and Safety

### 4.1 Controlled procedure

1. Prepare a stable dry work area. Keep laptops and displays away from the bath.
2. Build the assigned insulation and check the jar with cool water.
3. Prepare the ice-water bath and record its actual temperature.
4. Add 100 mL of water near 80 °C and record the actual starting temperature.
5. Centre the probe, close the neck, immerse the jar to the agreed depth, and start the timer.
6. Record the core temperature at  $t = 0$  and once per minute through  $t = 15$  min.
7. Enter the readings without smoothing them. Calculate drop, residuals, MAE, and an effective  $k$ .
8. Compare conditions and list the factors that limit the conclusion.

### 4.2 Fair-test criteria

Every condition should use the same jar geometry, water volume, probe depth, immersion depth, bath location, and timing method. The bath temperature must be measured rather than assumed. If a reading is late, the actual time should be recorded. A complete run contains 16 ordered time-temperature pairs or a written reason for each missing point.

### 4.3 Safety controls

| Hazard                 | Control                                                |
|------------------------|--------------------------------------------------------|
| Hot water              | Adult or trained supervisor pours; use gloves or tongs |
| Glass breakage         | Inspect jars; use stable secondary containment         |
| Water near electronics | Keep displays, laptops, and connectors dry             |
| Leak or unstable jar   | Stop the run immediately                               |
| Wet floor              | Use towels and clean spills at once                    |

### 4.4 Data quality

We kept the original readings, including the unexpected Mylar-only curve. The purpose of the model is to support a comparison, not to force a match. Repeating each configuration at least three times and randomising the run order would improve the evidence. Future trials should also record water mass, bath temperature, probe depth, immersion depth, and construction notes for every run.

## 5 Software and Data Workflow

### 5.1 Architecture

Mission Control is a static single-page application built with HTML, CSS, and JavaScript. It runs in a normal browser. Chart.js draws the curves and KaTeX renders equations. Local browser storage keeps the current prediction and telemetry during a session. No database, user account, native installation, Bluetooth link, or device driver is required.

### 5.2 Workflow

| Stage       | Website action                                            | Student action                            |
|-------------|-----------------------------------------------------------|-------------------------------------------|
| Prediction  | Calculates $T(t)$ from $T_0$ , $T_{env}$ , $k$ , and time | Choose a condition and record assumptions |
| Measurement | Provides timer and 16-point log                           | Read and enter one value per minute       |
| Comparison  | Overlays measured and predicted curves                    | Inspect curve shape and residuals         |
| Evaluation  | Calculates MAE and feedback score                         | Explain the disagreement physically       |

### 5.3 Graphing choices

The prediction and telemetry use the same time axis and configuration colours. Axes include units, and the plotted interval matches the 15-minute procedure. The data table keeps the measured value separate from the predicted value. This makes it clear which numbers came from the physical test and which came from the model.

The residual at each time is the measured value minus the model value. We use mean absolute error as a simple summary:

$$MAE = (1/n) \sum_{i=1}^n |T_{meas,i} - T_{model,i}|.$$

### 5.4 Reliability and limitations

A static site is easy to host and can be opened from a local server, but the current build still depends on browser libraries that may be loaded from external sources. A later version should bundle those files locally for dependable offline lessons. CSV export and run metadata would also make repeated tests easier to compare. The dashboard score is classroom feedback, not a formal statistical validation result.

### 5.5 Responsibility

Piyath Vithanage developed the website and the graphing and data-visualisation workflow. The thermal assumptions and experimental settings were coordinated with Dilan Fernando so that the software and physical procedure used the same definitions.

## 6 Experimental Results

### 6.1 Recorded endpoints

The worked dataset contains 16 readings for each condition. The endpoint summary is shown below. Because starting temperatures were not identical, final temperature alone is not enough for a fair comparison; the temperature drop and full curve are more useful.

| Condition    | T0      | T15     | Drop    | Dashboard score |
|--------------|---------|---------|---------|-----------------|
| Full MLI     | 77.6 °C | 62.7 °C | 14.9 °C | 93%             |
| Bubble wrap  | 82.6 °C | 52.6 °C | 30.0 °C | 83%             |
| Bare control | 80.1 °C | 35.2 °C | 44.9 °C | 94%             |
| Mylar only   | 80.5 °C | 33.3 °C | 47.2 °C | 3%              |

### 6.2 Main result

The full multilayer package had the smallest 15-minute temperature drop and therefore retained the most heat in this run. Bubble wrap ranked second. The bare control and Mylar-only condition cooled much faster. The Mylar-only result does not show that reflective insulation is generally ineffective. It shows that the selected reference coefficient and the physical water-bath construction did not agree for that run.

### 6.3 Representative telemetry

| Time   | Full MLI | Bubble | Mylar | Bare |
|--------|----------|--------|-------|------|
| 0 min  | 77.6     | 82.6   | 80.5  | 80.1 |
| 3 min  | 75.6     | 72.5   | 55.8  | 59.9 |
| 6 min  | 73.9     | 65.6   | 44.9  | 47.1 |
| 9 min  | 70.6     | 60.6   | 38.9  | 40.9 |
| 12 min | 67.4     | 56.1   | 35.5  | 37.3 |
| 15 min | 62.7     | 52.6   | 33.3  | 35.2 |

### 6.4 Fitted cooling constant

A point estimate of  $k$  can be calculated when  $T(t)$ ,  $T_0$ , and  $T_{env}$  are known:

$$k = -(1/t) \ln([T(t) - T_{env}] / [T_0 - T_{env}]).$$

A fit using all usable points is better than a one-point estimate because it uses the curve shape and reduces the influence of one reading. Repeated trials are still needed before reporting material rankings as stable results.

## 7 Analysis and Model Limitations

### 7.1 Comparison with the model

The full multilayer package and bare control followed their selected reference curves more closely than the Mylar-only case. The model is useful because it gives the class a common baseline, but  $k$  is an effective system value. It changes when the jar, fill, bath, probe, closure, or layer contact changes. A coefficient chosen for one setup should not be presented as a fixed property of a material.

### 7.2 Likely error sources

| Source                       | Effect on result                 | Control                         |
|------------------------------|----------------------------------|---------------------------------|
| Probe touches glass          | Biased core reading              | Centre probe with a guide       |
| Bath warms                   | Changing boundary                | Measure before and during run   |
| Different start temperatures | Unfair final-temperature ranking | Use drop, theta, or fitted $k$  |
| Compressed layers            | Higher solid conduction          | Use repeatable light fastening  |
| Water reaches wrap           | Extra conductive path            | Improve closure and log wetting |
| Timing delay                 | Shifted data point               | Record actual timestamp         |

### 7.3 Limits of the analogy

A spacesuit operates in vacuum outside the garment, where there is no external air convection. The classroom jar is surrounded by water, so convection and conduction are strong. Real spacesuits also use an active liquid-cooling and ventilation garment. Our activity therefore teaches the method of thermal testing and model checking; it does not reproduce the full physics or certify the materials.

### 7.4 Confidence in the result

The recorded ordering is valid as a description of the worked runs: full MLI, bubble wrap, bare control, then Mylar-only by 15-minute temperature drop. Confidence in a general ranking is limited because the available dataset contains one series per condition and incomplete metadata for some construction details. We would not claim that the same ordering will always occur until repeated trials show the variation.

### 7.5 Changes for the next campaign

- Run at least three repeats per condition and randomise the order.
- Use a simple probe-position jig and a fixed immersion-depth mark.
- Record bath temperature at the start and end of every run.
- Weigh the water instead of relying only on nominal volume.
- Store construction notes and export all telemetry as CSV.

## 8 Educational Delivery and Feasibility

### 8.1 Classroom sequence

The lesson starts with a short briefing on heat transfer and the limits of the spacesuit analogy. Groups then predict which condition will retain the most heat, assemble or inspect their test article, run the 15-minute measurement, and compare the recorded curve with the model. The final discussion asks students to defend a ranking and explain at least one source of disagreement.

### 8.2 Supporting materials

The project includes a student manual, teacher manual, task sheet, printable flashcards, an illustrated procedure, and the Mission Control website. We used the same configuration names, variables, test duration, and core volume throughout. Short instructions are placed beside the action; longer explanations are kept in the manual. This keeps the live procedure readable while preserving the engineering reason behind each control.

### 8.3 Assessment

| Evidence             | What it checks                          |
|----------------------|-----------------------------------------|
| Prediction           | Reasoning from heat-transfer mechanisms |
| Raw data table       | Careful measurement and units           |
| Model comparison     | Use of $k$ , residuals, and MAE         |
| Evaluation paragraph | Limits, uncertainty, and fair testing   |
| Open response        | Explanation in the student's own words  |

### 8.4 Practical feasibility

The equipment is reusable and the software does not require a paid account. Exact cost depends on local sourcing, so the earlier EUR 35.20 estimate should remain a draft until quotes are checked. The main preparation load is cutting consistent insulation pieces, checking jars and probes, and preparing the bath. One teacher can supervise the activity if hot-water handling is controlled and the work area is arranged before the lesson.

### 8.5 Handover

A teacher should receive the full kit, printed safety checklist, spare insulation, towels, and a local copy of the website. The handover notes should state the measured jar volume, probe type, reference coefficients, and known limits. This information is more useful than a polished demonstration alone because it lets another class repeat the test and compare results.

## 9 Conclusions, Responsibilities, and References

### 9.1 Conclusion

We developed a complete classroom engineering activity that connects a physical thermal test with a simple mathematical model. The full multilayer package retained the most heat in the worked data, while the Mylar-only result showed the importance of boundary conditions and construction. The strongest part of the project is the full workflow: students make a prediction, control a test, keep raw data, compare the result with a model, and explain the mismatch.

### 9.2 Final recommendations

- Repeat all four conditions and report the variation, not only one curve.
- Bundle Chart.js and KaTeX locally for offline use.
- Add CSV export, run identifiers, and metadata fields.
- Use a probe guide and fixed immersion mark.
- Verify the final cost from current supplier quotes.
- Keep the vacuum-versus-water-bath limitation in every student briefing.

### 9.3 Documented responsibility split

| Member               | Contribution recorded in project sources                                    |
|----------------------|-----------------------------------------------------------------------------|
| Piyath Vithanage     | Website, graphing, and data visualisation                                   |
| Dilan Fernando       | Thermal model, experiment, analysis, safety, documentation, and integration |
| Elio Krollpfeiffer   | Team member; specific deliverables not recorded                             |
| Nuhansee Migelhewage | Team member; specific deliverables not recorded                             |

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Project sources: Mission Control index.html, app.js, and styles.css; student and teacher manuals; task sheet; flashcards; project procedure; and the recorded telemetry dataset.

# Individual Contribution

## Piyath Vithanage

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**Status:** Documented contribution

### Role and completed work

My main contribution was the design and implementation of the Mission Control website. I organised the project as a single-page browser application so students could move through the briefing, assembly, prediction, measurement, study material, and assessment without installing software. I used HTML for the page structure, CSS for the responsive layout, and JavaScript for the model and interactive controls.

I connected the configuration selector, starting temperature, bath temperature, duration, and cooling constant to a common application state. I implemented the 15-minute timer and the manual telemetry workflow. The page validates the entered values, stores the active data in the browser, and keeps the measured series separate from the calculated reference series.

### Graphing and analysis

I developed the graphing workflow so students could compare the measured and predicted temperatures on the same time axis. I kept the configuration names and colours consistent across the selector, summary, graph, and result table. I also implemented the residual and mean absolute error calculations. These outputs turn the graph into an analysis task instead of using it only as a visual display.

### Design decisions

I avoided a native app and Bluetooth data transfer because they would add setup and failure points in a classroom. The static site works on standard laptops and tablets and can be hosted without a database. I also added a print view so the selected settings and telemetry can be kept with the worksheet.

| Deliverable                       | Evidence                          |
|-----------------------------------|-----------------------------------|
| Website structure and interaction | index.html and app.js             |
| Responsive visual design          | styles.css                        |
| Prediction and telemetry graphs   | Chart workflow in Mission Control |
| Residual and MAE feedback         | Analysis output and result table  |
| Printable experiment record       | Website print layout              |

I coordinated the interface variables with the thermal model so the website used the same 100 mL core, 15-minute duration, environmental temperature, and four configuration names as the physical procedure.

# Individual Contribution

## Elio Krollpfeiffer

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**Status:** Contribution statement requires author confirmation

### Record available at report finalisation

I am listed as a member of the SPHERE project team. The project files supplied for this final report do not identify a specific task, deliverable, test, or document as my individual responsibility. To keep the submission accurate, this page does not assign work that is not supported by the available record.

### Information to confirm before submission

- The project task or work package I owned.
- The practical, software, research, writing, presentation, or testing work I completed.
- The files, pages, test records, meeting notes, or version history that show my work.
- A short reflection on one decision, problem, or lesson from my contribution.
- An approximate share of the team effort, if the course requires it.

### Contribution entry

| Field                                | Author to complete               |
|--------------------------------------|----------------------------------|
| Primary role                         | Not recorded in supplied sources |
| Completed tasks                      | Not recorded in supplied sources |
| Evidence                             | Not recorded in supplied sources |
| Technical decision or problem solved | Not recorded in supplied sources |
| Date confirmed                       | _____                            |

### Author declaration

I confirm that the final text added to this page is a fair description of my own work and does not claim another team member's contribution.

Signature: \_\_\_\_\_ Date: \_\_\_\_\_

This page must be completed and checked by Elio Krollpfeiffer before the report is submitted if an individual contribution narrative is required.

# Individual Contribution

## Dilan Fernando

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**Status:** Documented contribution

### Role and completed work

My main technical responsibility was the thermodynamic definition of the experiment and the interpretation of the physical results. I selected the lumped Newtonian cooling model because it gives students a manageable equation while keeping the relationship between thermal mass, environmental temperature, and cooling rate. I defined  $k$  as an effective system coefficient in  $\text{min}^{-1}$  and kept it separate from material thermal conductivity.

I set the final test conditions: a 100 mL water core, a starting temperature near 80 °C, a measured ice-water bath, and readings every minute for 15 minutes. I connected those settings to the apparatus and student procedure so the model and the physical experiment referred to the same system.

### Experiment and analysis

I planned the four-condition comparison and analysed the worked telemetry using temperature drop, residuals, mean absolute error, and the fitted cooling constant. I treated the unexpected Mylar-only curve as evidence of a model or construction mismatch. I identified likely effects from probe position, closure, wetting, layer compression, bath temperature, and unequal starting temperatures.

### Safety, documentation, and integration

I developed the safety controls for hot water, glass, spills, and electronics. I aligned the terminology and sequence across the student manual, teacher manual, task sheet, procedure, website content, and report. I also completed the remaining technical writing and project integration recorded in the source report.

| Deliverable           | Evidence                                    |
|-----------------------|---------------------------------------------|
| Thermodynamic model   | Model equations and coefficient definitions |
| Experimental settings | 100 mL, near 80 °C, 15-minute plan          |
| Results analysis      | Telemetry comparison and limitations        |
| Safety controls       | Teacher and procedure guidance              |
| Document integration  | Manuals, task sheet, procedure, report      |

My main lesson from the project was that the boundary and assembly details can dominate a simple material comparison. For that reason, I kept the conclusion limited to the tested system and recommended repeated trials before making a broader ranking.

# Individual Contribution

## Nuhansee Migelhewage

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**Status:** Contribution statement requires author confirmation

**Record available at report finalisation**

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| Evidence                             | Not recorded in supplied sources |
| Technical decision or problem solved | Not recorded in supplied sources |
| Date confirmed                       | _____                            |

**Author declaration**

I confirm that the final text added to this page is a fair description of my own work and does not claim another team member's contribution.

Signature: \_\_\_\_\_ Date: \_\_\_\_\_

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